Taking real phone calls with the AI Receptionist

How a call reaches the application, which phone number is which, how the business's MCP server supplies live data, and the exact steps to run it.

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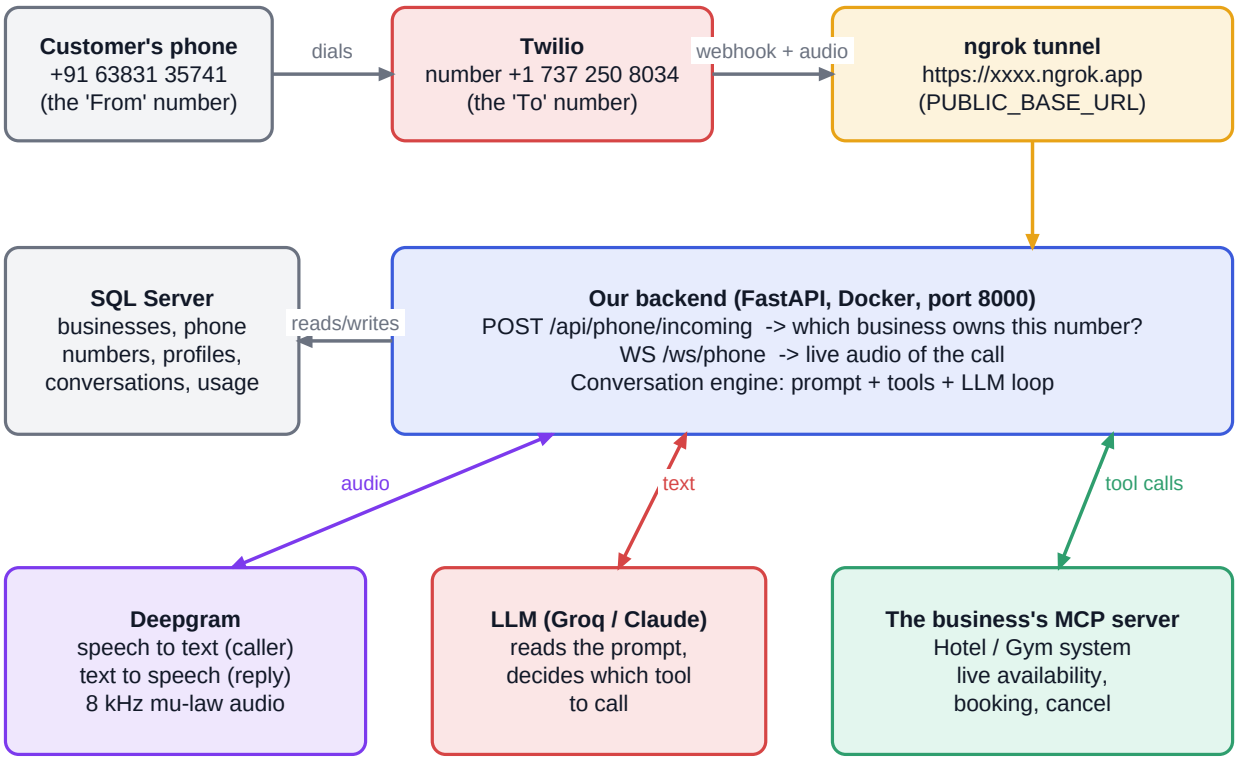
1. The big picture

A customer dials a phone number that belongs to a business. Twilio (the phone company layer) answers and forwards the call to our backend over the internet. Our backend turns the caller's speech into text, lets the AI decide what to say and which business tool to use, and speaks the answer back.



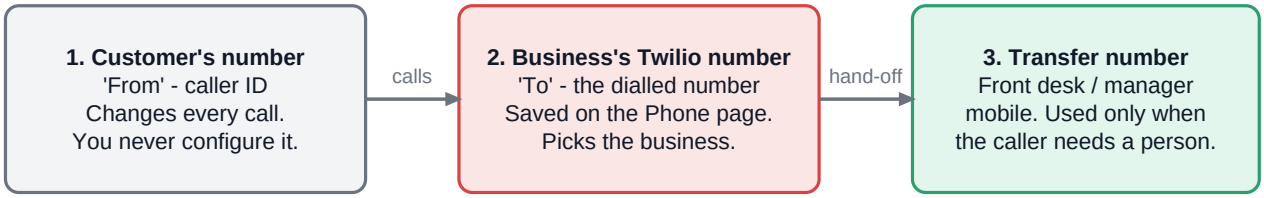
The caller only hears audio. Everything in the lower row happens in under a second, on every sentence.

The pieces

Piece	What it does	Where it lives
Twilio	Owns the phone number, answers calls, streams call audio to us, can transfer a call.	twilio.com (cloud)
ngrok	Gives your laptop a public https address so Twilio can reach it. Only needed in development.	your machine
Backend	Finds the business by the dialled number, runs the conversation, calls tools, records usage.	Docker, port 8000
Deepgram	Speech to text for the caller, text to speech for the reply.	deepgram.com (cloud)
LLM	Understands the request, chooses tools, writes the reply.	Groq / Claude (cloud)
MCP server	The business's own system: live availability, booking, cancellation.	the business's side

2. Which number is which

Three different phone numbers are involved. Mixing them up is the most common confusion.



How the backend picks the business:

Twilio sends To=+1 737 250 8034. The backend looks that number up in the phone_numbers table.

Sunrise Grand Hotel saved +1 737 250 8034 -> the call belongs to Sunrise Grand Hotel,

so the hotel's own profile, hours and MCP tools are used. Nobody else's data is reachable.

Numbers are unique: one number belongs to exactly one business at a time.

Where does a hotel's number come from?

The application does not pick a number by itself. **Each business needs its own Twilio number**, and the admin types it on the **Phone** page. There are three practical ways to get that number in front of customers:

Option	How	Best for
A. Publish the Twilio number	Buy a number in Twilio and print it on the hotel's website, Google listing and signage.	New lines, demos
B. Forward the hotel's existing number	The hotel keeps its public number and asks its phone provider to forward calls (always, when busy, or after hours) to the Twilio number. Customers notice nothing.	Most real hotels (recommended)
C. Move the hotel's number to Twilio	Port the existing number into Twilio. Takes days to weeks and needs paperwork.	Full production later

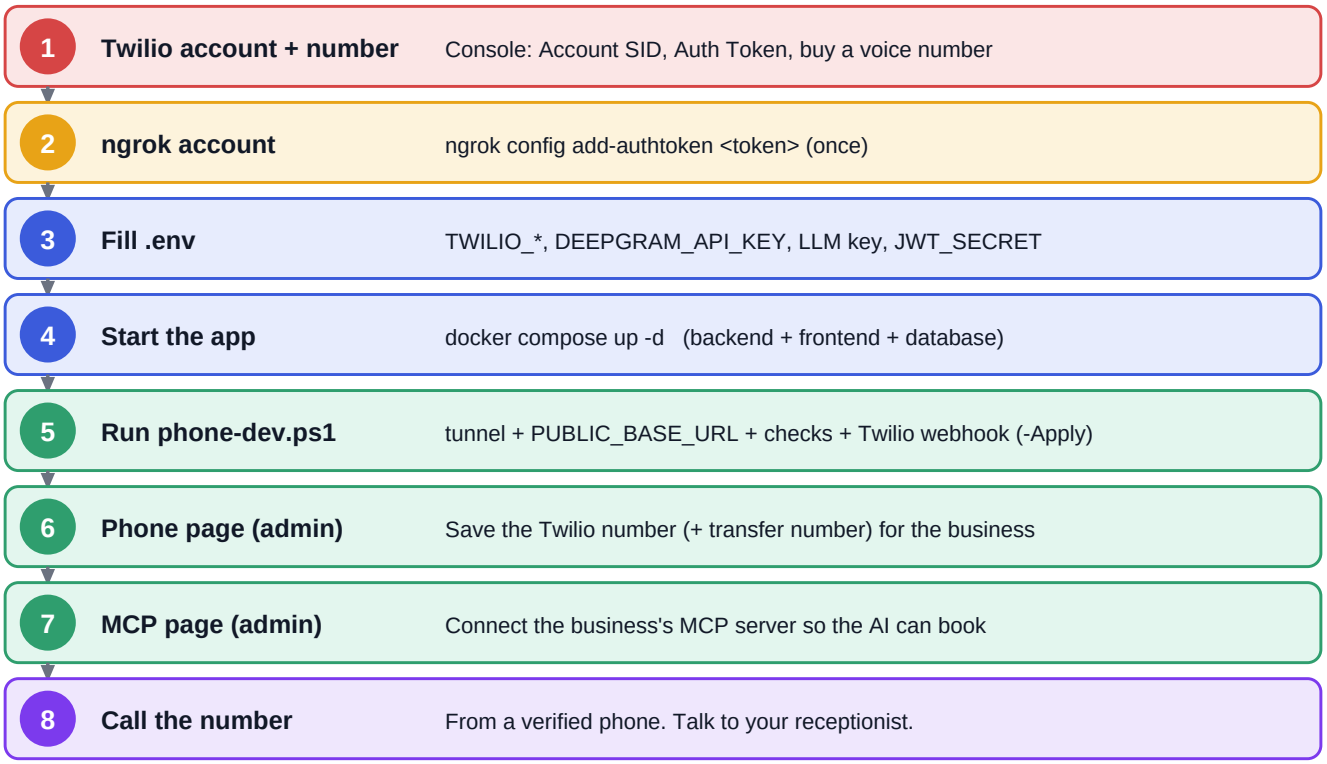
Example with two businesses

Business	Customers dial (Twilio number)	Transfer number (front desk)	MCP server
Sunrise Grand Hotel	+91 44 xxxx xxxx	+91 98xxx xxxxx (reception)	Hotel bookings
FitZone Anna Nagar	+91 44 yyyy yyyy	+91 97xxx xxxxx (club manager)	Gym bookings

Right now (trial account)

- You have **one** Twilio number: +1 737 250 8034. A number can belong to only one business, so for a demo save it on one business at a time. To switch from the hotel to the gym, clear it on the hotel's Phone page first (otherwise you get 'already used by another business'), then save it on the gym's.
- Callers: only phones verified in Twilio can call a trial number. Yours (+91 63831 35741) is verified.
- A second business needs a second Twilio number (paid account), or call forwarding as in option B.

3. Step-by-step setup and run



Commands

```
# once: let ngrok use your account
ngrok config add-authtoken <your-ngrok-token>

# every session, from the repo root
docker compose up -d
./scripts/phone-dev.ps1 -Number +17372508034 -Apply
```

What the script does: opens the tunnel, writes the tunnel URL into **PUBLIC_BASE_URL** in .env, restarts the backend, then checks your keys, that Twilio can reach the tunnel, and that the number's webhook points at `<tunnel>/api/phone/incoming`. With -Apply it sets that webhook for you. Every line it prints is [ok] or [FAIL]; fix any FAIL and run it again. A free ngrok URL changes each time ngrok restarts, so re-run it then.

Steps 6 and 7 in the application

Open the web app (http://localhost:5173) and log in as the **business admin** (not the superadmin).

Phone page

- **Business phone number**: +17372508034 (the Twilio number customers dial).
- **Transfer to a person**: the front desk mobile in the same +country-code form. Optional; without it the assistant promises a callback instead of transferring.
- Save. The page also shows the webhook URL (for checking only; -Apply already set it).

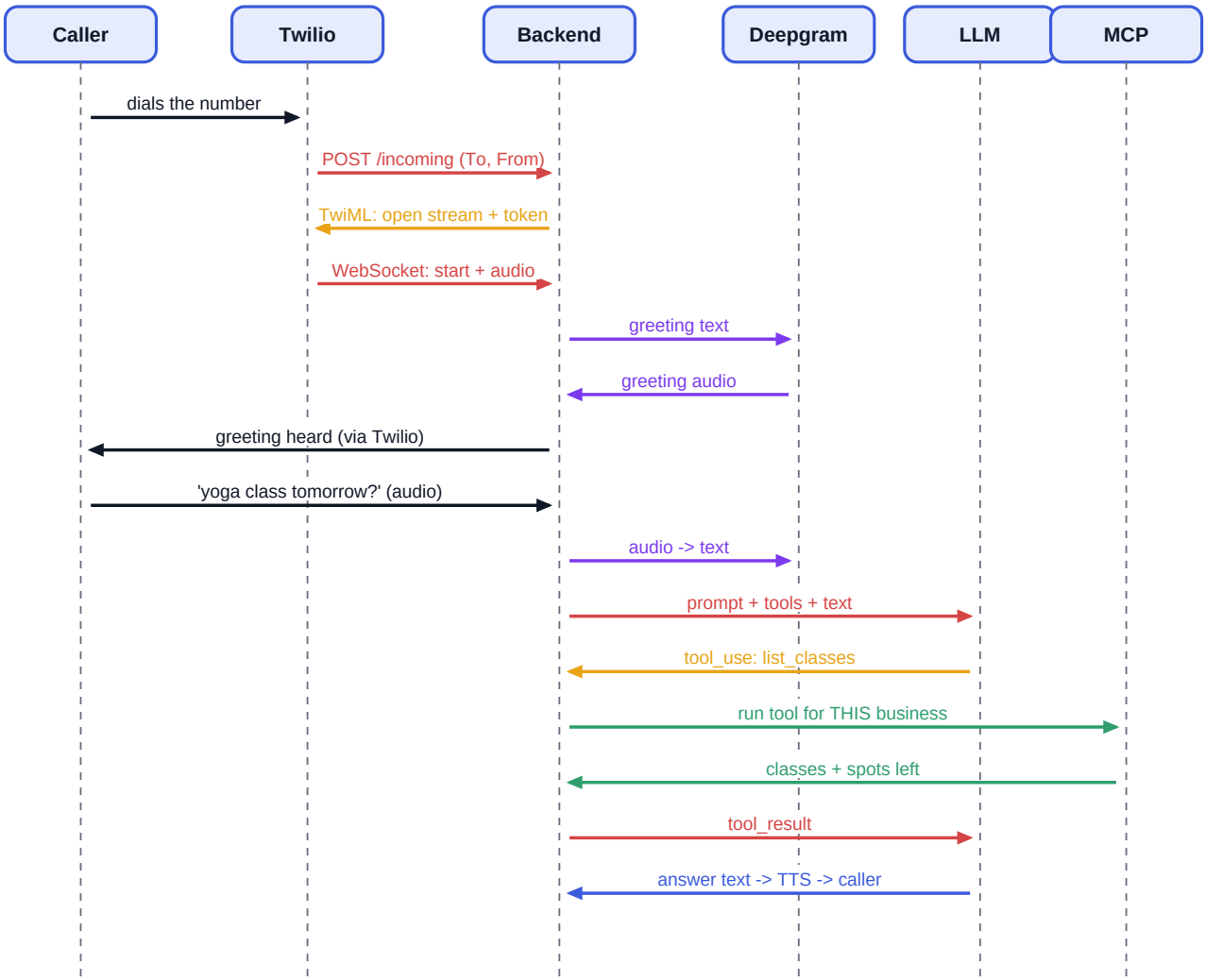
MCP integration page (so the AI can do real work)

- **Name**: for example 'Gym bookings'.
- **Server URL**: when the app and the MCP server both run in Docker on your PC use `http://host.docker.internal:8201/mcp` (the GymBooking MCP port). Not 'localhost': inside Docker that means the backend container itself.
- **Header**: click '+ API key' and enter name `X-API-Key` with a value from the GymBooking `GB_MCP_API_KEYS` setting.
- Transport: Streamable HTTP. Add the server; it lists the tools it found (list_gyms, list_classes, book_class ...).

Test before calling

- Chat & voice page, type: *'Do you have a yoga class tomorrow in Adyar?'*. If it answers with real classes, the MCP connection works and the phone will too.
- Then call the Twilio number from your verified phone and ask the same thing.

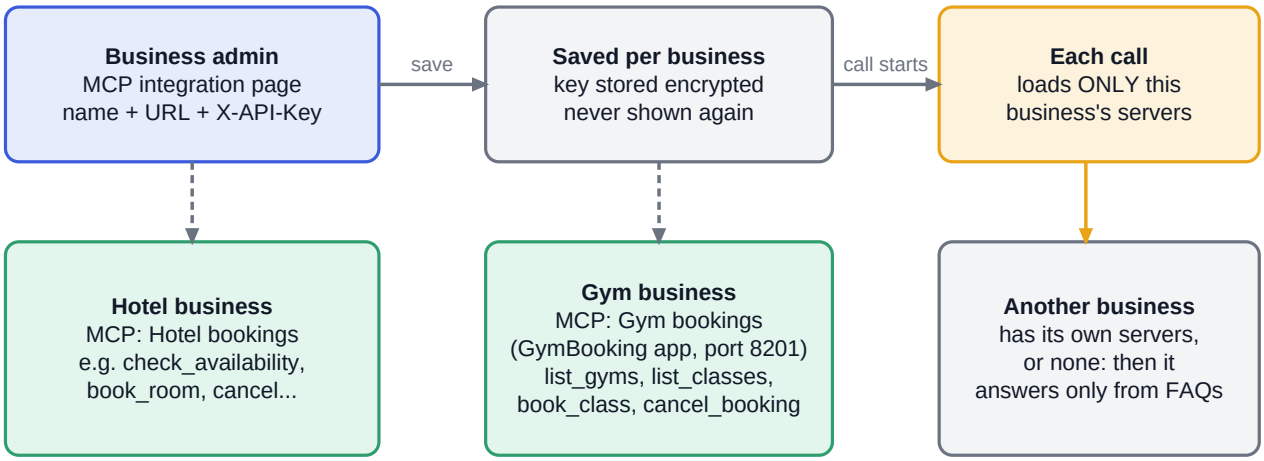
4. What happens during one call



Notes

- **Security:** Twilio's request is checked against its signature; the audio stream is allowed only with a 60-second signed token issued by the webhook, so nobody can open a stream for someone else's conversation.
- **Interruptions:** if the caller talks while the assistant speaks, the queued audio is cleared.
- **Person needed:** if the assistant calls request_human (or flags an urgent emergency) and the business is open with a transfer number set, the call is handed over after the assistant finishes its sentence.
- **Usage:** the call counts as a conversation, its messages, and its minutes on the dashboard.

5. How the business's MCP works on a call



A call to the hotel's number can never use the gym's tools, and the reverse: tools come only from the business that owns the dialled number. Platform tools (request_human, send_confirmation, flag_urgency) are added for every business next to its own MCP tools.

Example: a caller to the gym's number

Caller says	What the AI does	Tool / source
'Any yoga class tomorrow in Adyar?'	Looks up classes, reads back the first few with spots left.	MCP: list_classes
'Book the 7 AM one for Priya, priya@example.com'	Confirms the details, then books. Counted as a booking on the dashboard.	MCP: book_class
'Text me the confirmation'	Sends an SMS to the caller's number (caller ID is offered first).	Platform: send_confirmation
'What are your opening hours?'	Answers from the business's FAQs and hours; says it is closed after hours.	Knowledge base
'Let me talk to someone'	Says it is connecting them and transfers the call.	Platform: request_human

Without an MCP server the assistant can only answer from the FAQs, which is why a brand new business replies 'I don't have that information' to booking requests. Connect the MCP server first.

6. Configuration reference (.env)

Variable	Value / where to get it	Needed for
TWILIO_ACCOUNT_SID	Twilio console home, Account Info, starts with AC	Calls
TWILIO_AUTH_TOKEN	Same panel (click the eye). Secret: never paste it in chat or commit it.	Calls
TWILIO_SMS_FROM	+17372508034 (your Twilio number)	SMS confirmations
TWILIO_WHATSAPP_FROM	+17372508034 (sandbox; first send 'join twilio-trial' from your phone)	WhatsApp confirmations
PUBLIC_BASE_URL	Written by phone-dev.ps1 from the ngrok tunnel. Leave empty by hand.	Calls
DEEPGRAM_API_KEY	deepgram.com console (already in your .env)	Voice and calls
GROQ_API_KEY or ANTHROPIC_API_KEY	The LLM provider chosen by LLM_PROVIDER	Answers
JWT_SECRET	Any long random value; also signs the call's stream token	Security
MCP_ALLOW_PRIVATE_HOSTS / MCP_LOCALHOST_ALIAS	true / host.docker.internal for local MCP servers (already set)	Local MCP

Ports at a glance

Service	Port	Used by
AI Receptionist backend	8000	Twilio (via ngrok), the web app
AI Receptionist web app	5173	You, in the browser
GymBooking MCP server	8201	The backend, as http://host.docker.internal:8201/mcp
ngrok inspector	4040	See every webhook Twilio sent: http://127.0.0.1:4040

7. Trial limits and troubleshooting

Twilio trial account

- Only verified numbers can call in; callers hear a short trial announcement before the assistant answers.
- Free credit is limited (your console showed 100 messages and 69 voice minutes).
- WhatsApp sandbox: the tester must send 'join twilio-trial' first, and free-form messages work only within 24 hours of the tester's last message.

If something does not work

Symptom	Likely cause	Fix
Caller hears 'this number is not in service'	No business saved this number on its Phone page, or the business is disabled	Save the exact Twilio number (with +) on the admin's Phone page
Call rings then drops; ngrok inspector shows 403	Signature check failed: PUBLIC_BASE_URL differs from the URL Twilio called, or wrong auth token	Re-run phone-dev.ps1 (it resets the URL); check TWILIO_AUTH_TOKEN
Twilio says 'application error' / nothing happens	Tunnel down, backend not running, or the webhook still points at an old ngrok URL	Run phone-dev.ps1 -Number ... -Apply and read the [FAIL] lines
Greeting plays but it never hears you	DEEPGRAM_API_KEY missing or invalid	Check the key; backend log shows the Deepgram error
Answers 'I don't have that information'	No MCP server connected for that business and nothing in its FAQs	Connect the MCP server (section 3); test it in Chat first
MCP page says the server is unreachable	URL uses localhost, or MCP server not running, or wrong X-API-Key	Use host.docker.internal; start the GymBooking stack; check the key
'Already used by another business' on save	The number is saved on a different business	Clear it there first, then save it here
Asked for a person but not transferred	No transfer number saved, or the business is closed by its hours	Save a transfer number; check the hours setting

Where to look: **docker compose logs -f backend** for the app, **http://127.0.0.1:4040** for the requests Twilio sent, and the Twilio console under **Monitor, Logs, Calls** for what Twilio saw.